

Student ID 1 & 2: ~~(Do this in pairs)~~

Edit Distance rules for this worksheet:

- You are given two strings A and B .
- Allowed operations are insert, delete, and replace (each has cost 1).
- The goal is to transform A into B using minimum total cost.

① $k \rightarrow s$ ② $e \rightarrow i$

③ Insert g

itten

~~D k, I s~~

1. Warm-up: compute the distance

For each pair (A, B) , write the minimum edit distance.

}

A	B	ED(A, B)
kitten $\rightarrow$ sitting	sitting	3
flaw	lawn	
algorithm	altruistic	
care	score	

2. Choose a DP state

Let $m = |A|$, $n = |B|$, and define a table $ED[0..m][0..n]$. Complete the state definition: $ED[i][j] =$ _____ $ED[m][n] = \text{Edit Distance}(A, B).$

3. Fill an Edit Distance table

Use $A = \text{care}$ and $B = \text{score}$. Fill the full DP table $ED[i][j]$.

	ϵ	s	c	o	r	e
ϵ	0	1	2	3	4	5
c	1	①	1			
a	2					
r	3					
e	4					

$$1 + \min \left(\begin{array}{l} ED[i-1, j], \\ ED[i, j-1], \\ ED[i-1, j-1] \end{array} \right)$$

$$ED(i)(2) = ED(1, 2) = ED(0, 1)$$

4. Base cases

What are the base cases $ED[i][0]$ and $ED[0][j]$?

i j

I / D

5. Write the recurrence

Write the full recurrence for $ED[i][j]$ for both cases: $A[i] = B[j]$ and $A[i] \neq B[j]$.

$$ED[i][j] = \begin{cases} ED[i-1, j-1] & A[i] = A[j] \\ \text{else.} & \end{cases} \quad (*)$$

6. Complexity

What is the overall time and space complexity of DP based on above recurrence?

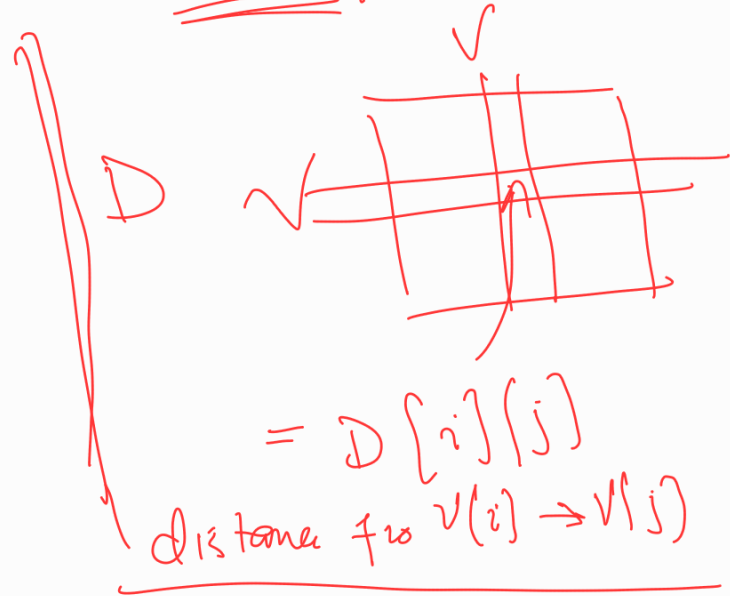
$ED \leftarrow \phi$
 for $i = 0 \rightarrow n$:
 $ED[i, 0] \leftarrow i$
 for $j = 0 \rightarrow m$:
 $ED[0, j] \leftarrow j$
 for $i = 1 \rightarrow n$:
 for $j = 1 \rightarrow m$:
 $ED[i, j] =$
 $ED[i-1, j-1]$
 if $A[i] == B[j]$
 else
 $ED[i, j] = 1 + \min(\text{---})$
 $\times \times$

Given $G = (V, E, \underline{W})$, find S.P.
 between every pair of nodes in V . distance
 $W: E \rightarrow \mathbb{R}$

Goal: Design a DP.

↳ Define state/memo.

Output



D

find shortest path $i \rightarrow j$

using only a subset of vertices as intermediate nodes.

(2)

$D[i, j] =$ distance $i \rightarrow j$

that only uses first k vertices.

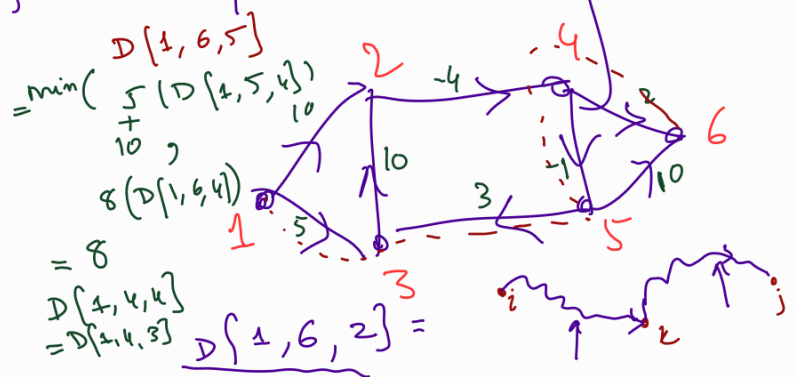
$D[i, j, k] = w[i, j]$

$\rightarrow \infty$

if $i, j \in E$ $k=0$

Initialization steps

$D[1, 1]$
 $D[2, 3]$



$$\underline{D[i, j, 1]} = D[i, j, 0]$$

distance from $i \rightarrow j$ that ~~may~~
use vertex $v[i]$ as someone
on the path.

$$\underline{D[i, j, n]}$$

use any of the first n
vertices

Recurrence $\min$ (

$$\underline{D[i, j, k]} = \min \left(\begin{array}{l} D[i, k, k-1] + D[k, j, k-1] \quad \textcircled{1} \text{ use vertex } k \\ D[i, j, k-1] \quad \textcircled{2} \text{ do not use vertex } k. \end{array} \right)$$

any of the
 $v[1 \dots k]$